

Computational Complexity, Homework Problem: Proof of Theorem 14.12

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Problem 4: On input, we are given an integer n and an undirected graph G with vertex set $V(G) = \{1, \dots, n\} \times \{1, \dots, n\}$ with the following properties:

- For every $1 \leq b \leq n$, there is no edge between vertices $\{(a, b) \mid 1 \leq a \leq n\}$.
- For every $1 \leq b, b' \leq n$, $b \neq b'$, and $1 \leq a \leq n$, there are at most $\log_2 n$ integers $1 \leq a' \leq n$ such that $(a, b)(a', b')$ is an edge of G .

We ask if G contains a clique of size n (i.e., a set of n pairwise adjacent vertices; note that they need to be of the form $(a_1, 1), (a_2, 2), \dots, (a_n, n)$ for some $1 \leq a_1, a_2, \dots, a_n \leq n$).

Prove that if there exists a constant $c > 1$ and an algorithm solving the problem above in time $O(c^n)$, then the Exponential Time Hypothesis fails.

Solution:

It is known from the lectures, that a $2^{o(n)}$ algorithm for 3-coloring would contradict the Exponential Time Hypothesis. Let the input be a graph G on n vertices. Assume that given problem Q is solvable in time $O(c^k)$ with respect to the number of vertex indices. Choose $k = O(\frac{n}{\log_3(\log_2(n))})$. We will show a reduction from 3-colorability of G to Q on k vertex indices.

This will prove that 3-colorability is solvable in $O(c^k) = O(2^{O(n \log_3^{-1}(\log_2(n)))}) = 2^{o(n)}$ time, which contradicts the ETH.

We proceed by dividing the vertices of G into k parts P_i each of size less than $\log_3(\log_2(n))$. We now insert all the possible 3-colorings of P_i into the corresponding column in the input graph of Q . We leave all the empty vertices unattended. Note that if some P_i doesn't have a valid coloring, then the whole graph doesn't have a valid 3-coloring, so we can return NO.

There are at most $3^{\log_3(\log_2(n))} = \log_2(n)$ possible colorings of P_i . Therefore, the number of non-empty vertices in each row is at most $\log_2(n)$, which will be important later.

We connect the non-empty vertices in the different rows, if those colorings are valid when G is restricted to the union of the those two sets of underlying vertices. Each non-empty vertex can have at most $\log_2(n)$ neighbors in each row, since there are at most $\log_2(n)$ non-empty vertices in each row.

We now claim that the graph G has a 3-coloring if and only if the answer to Q is "YES". Indeed, if G has a 3-coloring, then if we look at the colorings of the restrictions, they have to be valid when combined, hence there has to be a clique.

If there is a clique in Q , then all the colorings in the clique are valid when combined with each other. Therefore their total union has to be valid as well, hence G has a 3-coloring. The reduction is in polynomial time, therefore it is accounted for.